

Autonomous Obstacle Avoidance Robot

An autonomous robot simulation using sensor-based reactive AI for real-time obstacle avoidance. Built with Webots simulator and the e-puck robot platform, controlled by a Python script implementing proportional control inspired by Braitenberg vehicles.

Features

- Reactive AI with no pre-programmed paths or maps
- Proportional speed control for smooth curved avoidance paths
- Five-situation classification (CLEAR, FRONT_BLOCKED, LEFT_BLOCKED, RIGHT_BLOCKED, NARROW_PASSAGE)
- Stuck detection with automatic escape manoeuvre
- Speed smoothing via first-order low-pass filter
- Turn commitment to prevent oscillation
- Real-time path trail visualisation using Supervisor API

Requirements

- [Webots R2023b](#) (or later)
- Python 3.x (bundled with Webots)

Setup & Running

1. Open Webots
2. File → Open World → select `worlds/obstacle_avoidance.wbt`
3. Press Play to start the simulation
4. Observe the robot navigating the arena in the 3D viewport
5. Monitor console output for sensor readings and decisions

Project Structure

```
├── controllers/
│   └── obstacle_avoidance/
│       └── obstacle_avoidance.py      # Main controller script
├── worlds/
│   └── obstacle_avoidance.wbt        # Webots world definition
├── Pictures/
│   ├── robot_system_architecture_Extra.png
│   └── robot_sensor_layout_light.png
├── Test Results/                     # Screenshots and console logs
├── Project_Report.tex                # LaTeX report source
├── Project_Report.pdf                # Compiled report
└── README.md
```

Key Parameters

Parameter	Value	Description
SENSOR_THRESHOLD	80.0	Detection threshold (must exceed noise floor of ~76)
BASE_SPEED	3.14 rad/s	Normal cruising speed (50% of max)
MAX_SPEED	6.28 rad/s	Maximum motor velocity
STUCK_THRESHOLD	30 steps	Time before escape triggers (~2 seconds)
SMOOTHING_FACTOR	0.7	Speed smoothing (0=instant, 1=no change)

Results

- **100% collision avoidance** across all six test scenarios
- **320 seconds** of sustained autonomous operation
- **64 ms** detection-to-response latency
- Smooth curved navigation paths confirmed by trail visualisation



References

- Braitenberg, V. (1984). *Vehicles: Experiments in Synthetic Psychology*. MIT Press.
- Brooks, R.A. (1986). A robust layered control system for a mobile robot. *IEEE Journal on Robotics and Automation*, 2(1).
- Cyberbotics Ltd. (2023). *Webots User Guide*. <https://cyberbotics.com/doc/guide/index>
- Mondada, F. et al. (2009). The e-puck, a robot designed for education in engineering.